

Week 1.

Examples, Data, Models, Statistics, Empirical CDF, Quantiles, Simpson's paradox

1 Some Examples

Statistical inference concerns learning from experience: we observe a random sample $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and wish to infer properties of the distribution our data come from. Probability theory goes in the opposite direction: from the distribution we deduce the properties of a random sample $\mathbf{x}$, and of statistics calculated from $\mathbf{x}$. Statistical inference as a mathematical science has been developed almost exclusively in terms of probability theory.

Probability vs. statistics

Probability Previous studies showed that the drug was 80% effective. Then we can anticipate that for a study on 100 patients, in average 80 will be cured and at least 65 will be cured with 99.99% chances.

Statistics Observe that 78/100 patients were cured. We (will be able to) conclude that we are 95% confident that for other studies the drug will be effective on between 69.88% and 86.11% of patients

Example 1.1. M&Ms are sweets that come in k equally frequent colours. Suppose we do not know k . We sequentially examine 3 M&Ms and they are red, green, red. What would be your guess for the number of colors? Now suppose a fourth M&M is drawn and it is orange. What would be your guess now?

Example 1.2. Physicians Health Study was a 5-year randomized study published testing whether regular intake of aspirin reduces mortality from cardiovascular disease. Participants were male physicians 40-84 years old in 1982 with no prior history of heart attack, stroke, and cancer, no current liver or renal disease, no contraindication of aspirin, no current use of aspirin. Every other day, the male physicians participating in the study took either one aspirin tablet or a placebo. Response: whether the participant had a heart attack (including fatal or non-fatal) during the 5 year period. ¹

Condition	Heart attack	No heart attack
Aspirin	104	10933
Placebo	189	10845

Question: Is there an association between taking aspirin and having a heart attack (for men)?

¹Source: Preliminary Report: Findings from the Aspirin Component of the Ongoing Physicians Health Study. New Engl. J. Med., 318: 262-64, 1988

Example 1.3.

	Women			Men		
	applied	accepted	%	applied	accepted	%
Computer Science	26	7	27	228	58	25
Economics	240	63	26	512	112	22
Engineering	164	52	32	972	252	26
Medicine	416	99	24	578	140	24
Veterinary Medicine	338	53	16	180	22	12
Total	1184	274	23	2470	584	24

Figure 1: The Simpson paradox for sex bias.

Example 1.4.

	Treatment A			Treatment B		
	treatment	success	%	treatment	success	%
Small stones	87	81	93	270	234	87
Large stones	263	192	73	80	55	69
Total	350	273	78	350	289	83

Figure 2: The Simpson paradox for kidney surgeries.

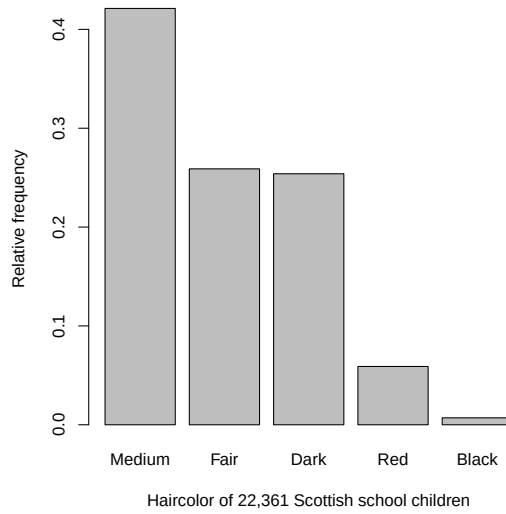
2 Data

Definition 2.1. (Data) $(x_1, \dots, x_n)$, $x_i \in S, i = 1, \dots, n$. S is usually $\mathbb{R}, \mathbb{R}^d$ or it can be any abstract set. But for us S (the sample space) will usually be $\mathbb{R}$.

Data can be *qualitative* (or *categorical*) or *numerical* (or *measurement*). Numerical data can be *discrete* or *continuous*.

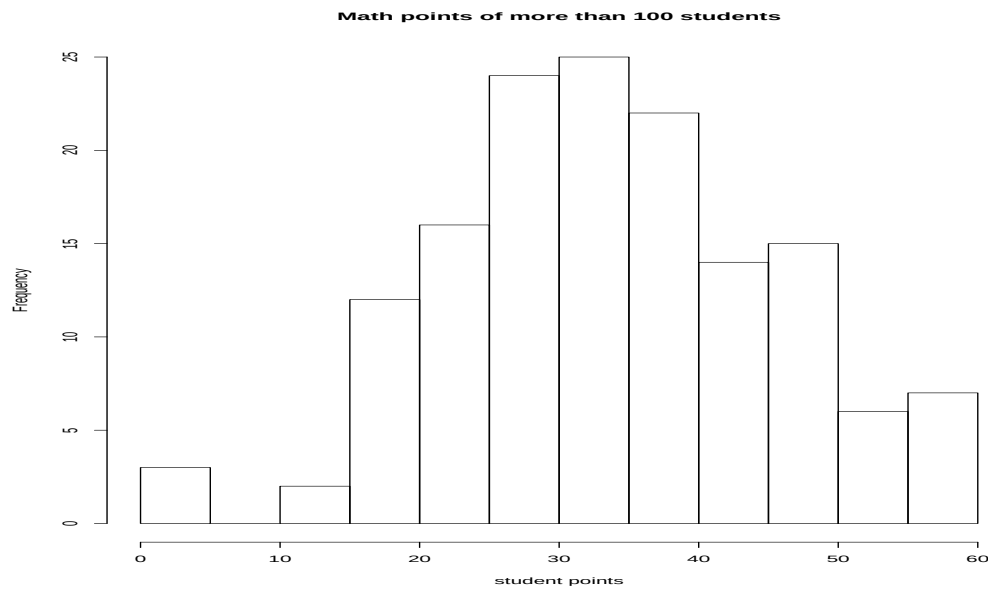
Categorical: hair color, blood type. Numerical: height of a person (of which distribution is approximately normal), number of daily accidents (of which distribution is approximately Poisson).

We can plot the frequencies of discrete or categorical data in a line diagram:



The total of the relative frequencies is clearly 1.

The continuous data can be summarized by a histogram:



The total of the rectangle areas is 1.

3 Statistical Model

Definition 3.1. (Sample) In statistics, our data are often modeled by a vector of IID (independent, identically distributed) random variables $X = (X_1, X_2, \dots, X_n)$ (called the sample, of which size is n), where the X_i take values in $\mathbb{Z}$ or $\mathbb{R}$.

Definition 3.2. A statistical (parametric) model is a family $\{\mathbb{P}_\theta \mid \theta \in \Theta\}$ of distributions on the sample space. When $\Theta \subset \mathbb{R}^d$ we say that we have a parametric model and we call Θ the parameter space.

If $X \sim N(\mu, \sigma^2)$ then we write

$$f(x \mid \mu, \sigma) = \frac{1}{\sigma(2\pi)^{1/2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

for its probability density function (PDF) indicating that the PDF depends on μ and σ . In this case we can model the problem as $\{f(x \mid \mu, \sigma) \mid (\mu, \sigma) \in \mathbb{R} \times \mathbb{R}^+\}$ (since the PDF determines the distribution), or, if $X = (X_1, X_2, \dots, X_n)$, $X_i \sim N(\mu, \sigma^2)$ the X_i being independent, we write

$$f(x \mid \mu, \sigma) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|x - \mu \mathbf{1}\|^2}{2\sigma^2}\right)$$

where $x = (x_1, \dots, x_n)$, $\mathbf{1} = \underbrace{(1, 1, \dots, 1)}_{\text{n times}}$.

In general, we write $f(x; \theta)$ or $f_\theta(x)$ or $f(x \mid \theta)$ for the probability density function of the probability mass function. The PDFs or PMFs (or, laws) form the **parametric family** $\{f(x \mid \theta) \mid \theta \in \Theta\}$. The statistician observes x from $f(x \mid \theta)$ and infers the value of θ .

4 The Empirical Distribution Function (EDF)

Definition 4.1. Having data $(x_1, x_2, \dots, x_n)$, we define the empirical distribution function $\hat{F}_n$ as

$$\hat{F}_n(x) = \frac{1}{n} \cdot \#\{i : x_i < x\}$$

Remark 4.2. The empirical distribution function is the distribution function of a random variable X that takes each of the x_i with probability $1/n$.

Proposition 4.3. If we have a sample $(X_1, \dots, X_n)$ with parent distribution function F then for every x we have the random variable $\hat{F}_n(x)$ and it is clear that $n\hat{F}_n(x) \sim \text{Bin}(n, F(x))$, so for every x we have

$$i) \mathbb{E}\hat{F}_n(x) = F(x);$$

$$ii) \text{Var}\hat{F}_n(x) = \frac{F(x)(1-F(x))}{n}$$

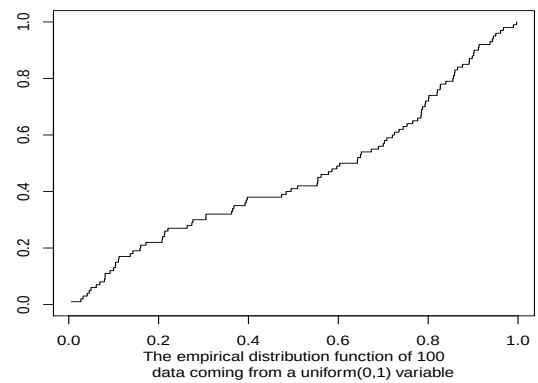
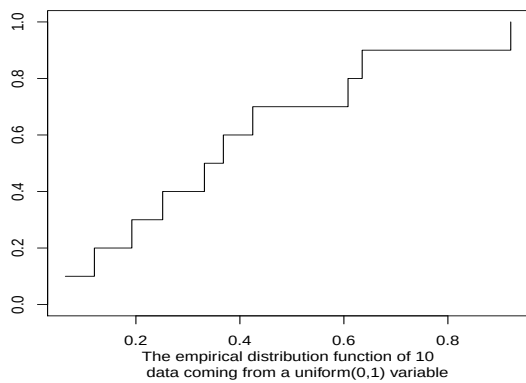
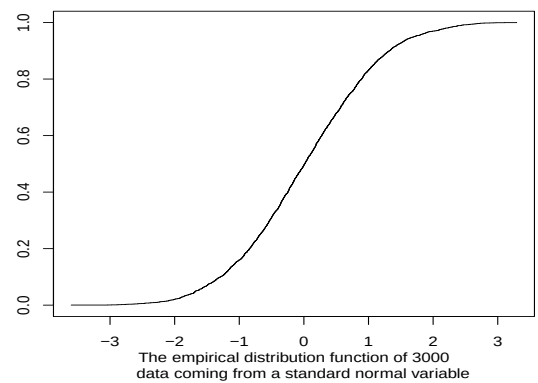
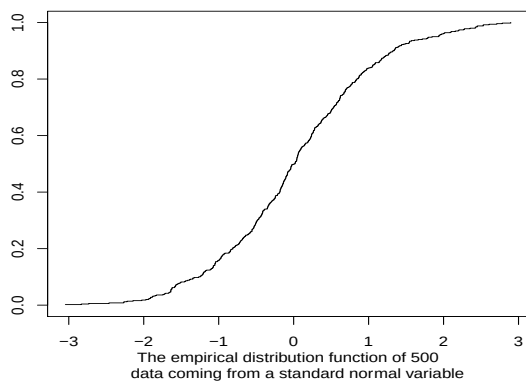
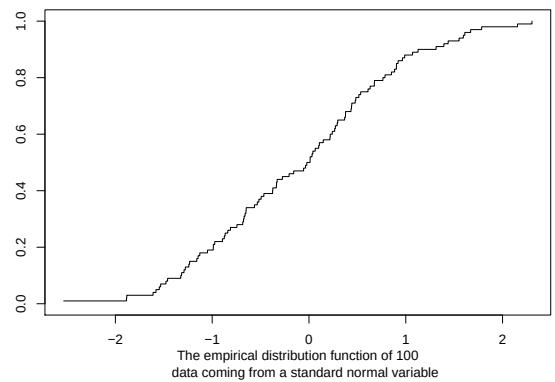
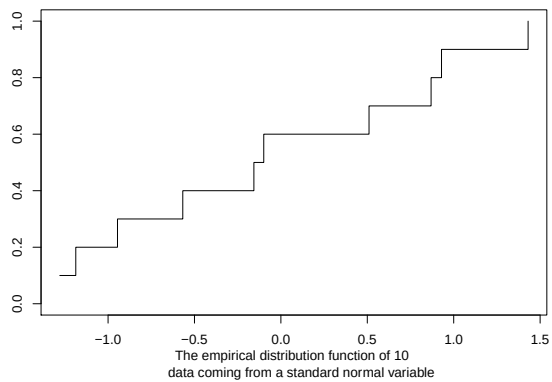
Moreover, by the SLLN we have

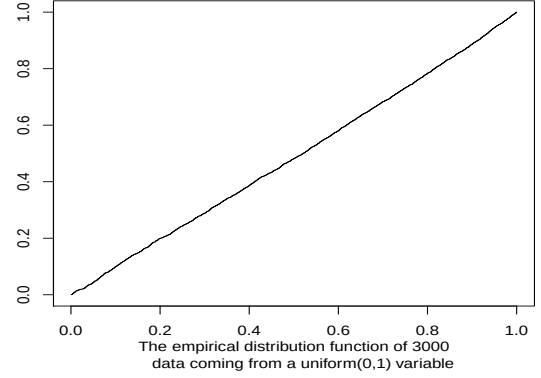
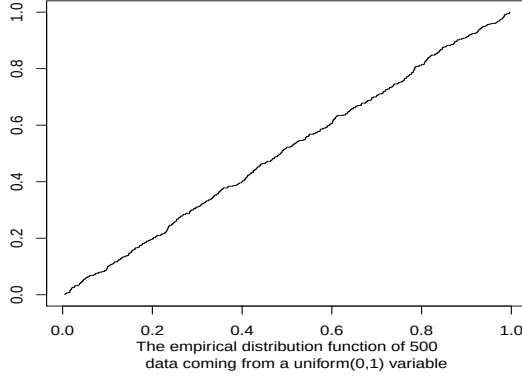
$$iii) \lim_{n \rightarrow \infty} \hat{F}_n(x) = F(x) \text{ with probability } 1.$$

But even more is true:

Theorem 4.4. (Glivenko-Cantelli) With probability 1

$$\lim_{n \rightarrow \infty} \sup_x |\hat{F}_n(x) - F(x)| = 0.$$





5 Statistics

Definition 5.1. (Statistic) A statistic is any function of the sample. For example, $\bar{X} := \frac{1}{n}(X_1 + \dots + X_n)$, $\max(X_1, \dots, X_n)$, $5(X_1, \dots, X_n)$, $S^2 := \frac{1}{n-1} \sum_{i=1}^n (\bar{X} - X_i)^2$ or $(\bar{X}, S^2)$ are statistics.

Definition 5.2. (Order statistic) If $X_1, X_2, \dots, X_n$ are IID rvs then the statistic

$$(X_{(1)}, X_{(2)}, \dots, X_{(n)}) \quad (1)$$

is called the order statistic where we put the same numbers in sorted order so $X_{(1)}$ is the least and $X_{(n)}$ is the greatest.

Example 5.3. Let the sample $(X_1, \dots, X_n)$ come from a uniform distribution on $(0, 1)$. Then $X_{(n)} \sim G(x)$, where $G(x) = \mathbb{P}(X_{(n)} < x) = \mathbb{P}(X_1 < x, \dots, X_n < x) = x^n$, $x \in (0, 1)$ and therefore its probability density function (PDF) is $g_{X_{(n)}}(x) = nx^{n-1}$, $x \in (0, 1)$. Later on we will need its expectation and variance:

$$\mathbb{E}X_{(n)} = \int_0^1 x \cdot nx^{n-1} dx = \frac{n}{n+1},$$

$$\text{Var}X_{(n)} = \mathbb{E}X_{(n)}^2 - (\mathbb{E}X_{(n)})^2 = \frac{n}{n+2} - \left(\frac{n}{n+1}\right)^2 = \frac{n}{(n+1)^2(n+2)}.$$

What is the law of $X_{(1)}$? Of $X_{(k)}$? Their expectations and variances?

6 Quantiles of a Random Variable

Definition 6.1. (p -th quantile of a rv) If X is a random variable and $0 < p < 1$, then the p -th quantile (or, the p -th percentile) of X (or, of the distribution of X) is any number x such that $\mathbb{P}(X \leq x) \geq p$ and $(\mathbb{P}(X \geq x) \geq 1 - p)$.

Remark 6.2. It is not hard to see that when p is in the range of the distribution function F (of X) then the set of p - quantiles is the closure of the (possibly degenerate) interval on which $F(x) = p$; and it is

$$\sup\{x \mid F(x) < p\} \tag{2}$$

when p is not in the range of F . In this case (due to the left continuity of F) we can also write $\max$ in (2) instead of $\sup$.

Exercise 6.3. *Prove the statements in Remark 6.2.*